

High-Frequency Market Making via Reinforcement Learning under Different Volatility Regimes

Technical Report Template Adaptation - thesis_31

Field	Value
Project	Synthetic HFMM reinforcement-learning thesis
Draft	manuscript/thesis_31.docx and manuscript/thesis_31.pdf
Frozen baseline	manuscript/thesis_29.pdf, tag thesis-v29-frozen
Source draft	manuscript/thesis_30.pdf / .docx
Decision log	manuscript/decisions_log_13.pdf
Scope	Template adaptation only; no experiment reruns or figure regeneration

In the tested controlled synthetic HFMM environment, explicit categorical volatility-regime labels do not provide robust incremental value once $\sigma_{\hat{}}$ is already observed by the PPO policy. The results are consistent with signal redundancy.

Abstract

This report studies whether explicit volatility-regime information improves a PPO-based high-frequency market-making agent in a controlled synthetic limit-order-book environment. The simulator uses an arithmetic Brownian-motion mid-price, a sticky Markov volatility-regime process, rolling realized-volatility detection, and Poisson-arrival fills. The PPO agent chooses bid and ask quotes through a discrete half-spread and skew parameterization, and it is compared with a fixed-spread strategy and an Avellaneda-Stoikov analytical baseline.

In the tested controlled synthetic HFMM environment, explicit categorical volatility-regime labels do not provide robust incremental value once σ_{hat} is already observed by the PPO policy. The results are consistent with signal redundancy. The evidence comes from the canonical WP5/WP6 experiments: out-of-sample evaluation, detector robustness, oracle-label ablations, regime-conditional reward shaping, mild model misspecification, and a signal-informativeness sweep. The strongest interpretation is therefore conditional and synthetic-market bounded: the explicit regime label mostly repackages volatility information already present in the continuous signal. The report makes no live-trading deployment claim and does not claim to prove the internal PPO mechanism.

Symbols, Abbreviations, and Glossary

Symbol	Meaning
S_t	Mid-price at time step t .
q_t	Inventory after step t .
X_t	Cash account after step t .
W_t	Marked-to-market wealth or equity, $W_t = X_t + q_t S_t$.
σ_{hat}	Rolling realized-volatility estimate observed by the PPO policy.
R_t	Reward at time t .
h	Quoted half-spread in ticks.
m	Quote skew in ticks.
A, k	Poisson fill-intensity scale and decay parameters.
η	Inventory penalty coefficient.

Abbreviation	Meaning
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ABM	Arithmetic Brownian motion.
AS	Avellaneda-Stoikov baseline.
HFMM	High-frequency market making.
OOS	Out-of-sample evaluation.
PPO	Proximal Policy Optimization.
RV	Realized volatility.
TOST	Two One-Sided Tests equivalence procedure.
WP	Work package.

Term	Definition
Regime-aware PPO	PPO policy variant that receives a categorical regime one-hot channel.
Regime-blind PPO	PPO policy variant that omits the regime one-hot but still observes $\sigma_{\hat{}}$.
σ_{only}	Ablation variant using the continuous volatility signal without categorical regime labels.
combined	Ablation variant using $\sigma_{\hat{}}$ and estimated categorical regime labels.
oracle_full	Ablation variant using $\sigma_{\hat{}}$ and the true regime label.

1 Introduction

1.1 Problem Statement and Motivation

Market makers continuously choose bid and ask quotes while balancing spread capture against inventory risk. When volatility changes, the same quote width can become either too passive or too exposed. A natural response is to provide a learning agent with regime labels such as low, medium, and high volatility. The central question is whether that categorical label adds decision-relevant information when the policy already observes a continuous volatility estimate.

1.2 Research Objective

The objective is to test the incremental value of explicit categorical volatility-regime labels in PPO market making under a controlled synthetic environment. The comparison is intentionally information-design focused: the regime label is evaluated against a continuous σ_{hat} signal, not against an observation space with no volatility information.

1.3 Background and Related Work

Classical market-making models, especially the Avellaneda-Stoikov framework and later inventory-risk extensions, formalize the tradeoff between spread income and inventory exposure. They provide analytical baselines and clarify why volatility and inventory jointly affect reservation prices and spreads.

Reinforcement-learning approaches to market making replace closed-form quoting rules with learned policies that can condition on state variables, simulated fills, and reward shaping. Prior DRL market-making work motivates the use of PPO-style policies, but it also raises a representation question: adding state channels can help only when they provide incremental information that the policy can exploit.

Volatility regimes are a common way to summarize non-stationarity. In this thesis, regimes are generated by a three-state Markov process and estimated by rolling realized volatility. The literature motivates regime awareness, but the tested design asks a narrower question: whether the categorical regime label improves performance beyond the continuous realized-volatility proxy.

Because the main empirical result is a bounded null/equivalence-style finding, the report uses paired tests and TOST equivalence tests rather than relying only on non-significant p-values. This supports a defense-safe reading of practical equivalence where the evidence warrants it.

1.4 Contributions

1. A controlled HFMM simulator with Poisson-arrival fills, fees, latency, and Markov volatility regimes.
2. A Gymnasium environment in which PPO controls quote half-spread and skew.
3. An OOS comparison of fixed-spread, Avellaneda-Stoikov, regime-aware PPO, and regime-blind PPO strategies.
4. A five-variant ablation separating continuous volatility information from estimated and oracle regime labels.
5. Detector, reward-shaping, misspecification, and signal-informativeness checks that bound the interpretation.

1.5 Scope and Limitations

The thesis is a controlled synthetic-market study. It does not claim live trading viability, real-order-book external validity, or a proven PPO representation mechanism. The result is conditional on the simulator, signal design, PPO hyperparameters, and degradation calibration tested here.

1.6 Report Structure

Sections 2-6 define the mathematical model, use case, architecture, layered model, and algorithms. Sections 7-9 describe the experimental setup, results, and metrics. Sections 10-12 discuss the interpretation, reproducibility, and conclusion. Appendices provide code maps, sanity checks, and extended evidence.

2 Mathematical Formulation

2.1 Synthetic Mid-Price Process

$$S_{t+1} = S_t + \sigma_t \sqrt{dt} \epsilon_t, \quad \epsilon_t \sim N(0, 1)$$

Here S_t is the mid-price, σ_t is the step volatility in ticks or price units according to the simulator configuration, dt is the time step, and ϵ_t is an independent standard-normal shock.

2.2 Markov Volatility-Regime Process

$$P(z_{t+1}=j \mid z_t=i) = P_{ij}, \quad z_t \in \{L, M, H\}$$

Here z_t is the latent volatility regime at step t , L/M/H denote low, medium, and high volatility, and P_{ij} is the sticky transition probability from regime i to regime j .

The regime controls the volatility multiplier applied to the base σ parameter. The canonical full experiments use σ multipliers [0.6, 1.0, 1.8] for L, M, and H.

2.3 Realized Volatility Signal and Regime Detection

$$\sigma_{\hat{t}} = \sqrt{(1/w) \sum_{i=t-w+1}^t (\Delta S_i)^2}$$

Here $\sigma_{\hat{t}}$ is the rolling realized-volatility signal, w is the rolling window length, and ΔS_i is the mid-price increment over step i .

Estimated regime labels are assigned by thresholding $\sigma_{\hat{t}}$ after a warmup period. The main WP4/WP5/WP6 pipelines use the causal $rv_baseline$ detector. The rv_dwell detector is retained only as an auxiliary/offline robustness comparison, while the HMM detector is an additional robustness variant.

2.4 Limit-Order Fill Model

$$\lambda(\delta) = A \exp(-k \delta)$$

Here $\lambda(\delta)$ is the fill intensity at quote distance δ , A is the baseline arrival scale, and k controls the exponential decay as quotes move farther from the mid-price.

$$P(\text{fill} \mid \delta) = 1 - \exp(-\lambda(\delta) dt)$$

Here $P(\text{fill} \mid \text{delta})$ is the per-step fill probability and dt is the simulator step length.

2.5 Quote Parameterization

$$h = h_{\text{idx}} + 1, \quad m = m_{\text{idx}} - 2$$

Here h_{idx} and m_{idx} are the two discrete PPO action components, h is the half-spread in ticks, and m is the quote skew in ticks.

$$\text{delta_bid} = \max(1, h + m), \quad \text{delta_ask} = \max(1, h - m)$$

Here delta_bid and delta_ask are the bid and ask quote distances in ticks. The max operator enforces a minimum quote distance of one tick.

2.6 Inventory, Cash, Equity, and Reward

$$q_{t+1} = q_t + F_t^{\text{bid}} - F_t^{\text{ask}}$$

Here q_t is inventory, F_t^{bid} is the bid-side fill indicator or count, and F_t^{ask} is the ask-side fill indicator or count.

$$X_{t+1} = X_t - F_t^{\text{bid}} P_t^{\text{bid}} + F_t^{\text{ask}} P_t^{\text{ask}} - \text{fees}_t$$

Here X_t is cash, P_t^{bid} and P_t^{ask} are the executed bid and ask prices, and fees_t denotes transaction costs.

$$W_t = X_t + q_t S_t$$

Here W_t is mark-to-market wealth or equity, X_t is cash, q_t is inventory, and S_t is the mid-price.

$$R_t = W_{t+1} - W_t - \eta q_{t+1}^2$$

Here R_t is the reward and η is the inventory-penalty coefficient. Fees are included in the cash update and are not counted a second time in the reward.

2.7 Avellaneda-Stoikov Baseline

$$r_t = S_t - q_t \gamma \sigma^2 \tau$$

Here r_t is the reservation price, γ is inventory-risk aversion, σ is volatility, and τ is remaining horizon.

$$\text{delta_AS} = 0.5 \gamma \sigma^2 \tau + (1 / \gamma) \log(1 + \gamma / k)$$

Here delta_AS is the AS half-spread and k is the same fill-intensity decay parameter used in the execution model. The implemented deltas are clipped to configured minimum and maximum bounds.

2.8 PPO Objective

$$\rho_t(\theta) = \pi_\theta(a_t \mid s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t)$$

Here $\rho_t(\theta)$ is the PPO probability ratio, π_θ is the current policy, $\pi_{\theta_{\text{old}}}$ is the behavior policy used to collect the rollout, a_t is the action, and s_t is the state.

$$L_{\text{CLIP}}(\theta) = E_t[\min(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t)]$$

Here θ denotes policy parameters, $\rho_t(\theta)$ is the PPO probability ratio, $A_{\hat{t}}$ is the advantage estimate, and ϵ is the clipping parameter. The report uses this only as a concise training-objective reference.

3 Use-Case Scenario

3.1 Synthetic HFMM Scenario

The use case is a synthetic market maker posting one bid and one ask quote at each step. The environment supplies mid-price dynamics, volatility estimates, and regime labels; the agent receives stochastic fills and is evaluated on wealth, risk-adjusted performance, inventory tails, and fill behavior.

3.2 Trading Agents and Strategy Variants

Strategy or variant	Observed information	Purpose
Naive	No learned state dependence	Fixed-spread baseline.
Avellaneda-Stoikov	Inventory, volatility, horizon	Analytical inventory-risk baseline.
ppo_aware	$\sigma_{\hat{t}}$ and estimated regime one-hot	Original regime-aware PPO.
ppo_blind	$\sigma_{\hat{t}}$ without regime one-hot	Original regime-blind PPO.
sigma_only	Continuous $\sigma_{\hat{t}}$ only	Tests whether $\sigma_{\hat{t}}$ alone carries the signal.
combined	$\sigma_{\hat{t}}$ and estimated regime one-hot	Tests estimated categorical incremental value.
oracle_full	$\sigma_{\hat{t}}$ and true regime one-hot	Tests perfect-label incremental value.
regime_only / oracle_pure	Categorical labels without $\sigma_{\hat{t}}$	Anchor label-only variants.

3.3 Information-Design Question

In the tested controlled synthetic HFMM environment, explicit categorical volatility-regime labels do not provide robust incremental value once $\sigma_{\hat{t}}$ is already observed by the PPO policy. The results are consistent with signal redundancy.

4 System Architecture

4.1 Run Lifecycle and Reproducibility Layer

``run.py`` dispatches jobs from JSON configuration files, creates a timestamped run directory, snapshots the config, records git metadata, writes logs and metrics, and finalizes run status. Long WP6 jobs support resume validation against the saved config snapshot.

4.2 Synthetic Market and Detector Layer

``src/wp1/sim.py`` implements the fill and inventory simulator. ``src/wp2/synth_regime.py`` generates synthetic regime paths, mid-prices, rolling realized volatility, and detector outputs.

4.3 Gymnasium Environment Layer

``src/wp3/env.py`` wraps the simulator as a Gymnasium environment. The observation vector is $[q_norm, \sigma_hat, \tau, regime_L, regime_M, regime_H]$. Warmup, invalid, or disabled regime labels produce a zero one-hot vector rather than an artificial medium label.

4.4 Strategy and Training Layer

WP4 trains PPO policies; WP5 evaluates strategies and ablations; WP6 runs the signal-informativeness sweep. The fixed-spread and AS baselines provide non-learning comparators.

4.5 Evaluation and Evidence Layer

Evaluation writes CSV metrics, figures, and summaries. The `thesis_31` adaptation embeds existing figures and quotes frozen numerical evidence; it does not regenerate evidence artifacts.

5 Layered Experimental Model

5.1 Layer 1: Market State Generation

Generate Markov regimes and mid-price paths under controlled volatility multipliers.

5.2 Layer 2: Signal Construction

Construct σ_hat from rolling realized volatility and derive estimated regime labels.

5.3 Layer 3: Observation Encoding

Encode inventory, volatility, time-to-horizon, and optional regime one-hot channels.

5.4 Layer 4: Action and Execution

Decode PPO actions into bid/ask quote distances and sample Poisson-arrival fills.

5.5 Layer 5: Learning and Evaluation

Train PPO on the chronological train segment and evaluate deterministic policies OOS.

6 Algorithm Specification and Pseudocode

6.1 Plain-Language Overview

The experiment creates a synthetic market path, estimates volatility signals, trains policies on the first 70% of the path, and evaluates all strategies on the held-out 30%. Ablations modify which volatility channels the PPO policy can observe.

6.2 Synthetic Market and Regime Generation

Input: seed, transition matrix P , base volatility, volatility multipliers, n_steps

Initialize z_0 and S_0

For $t = 0 \dots n_steps - 1$:

sample z_{t+1} from $P[z_t]$

set $\sigma_t = \text{base_sigma} * \text{multiplier}[z_t]$

sample mid-price increment and update S_{t+1}

Compute rolling σ_{hat} after warmup window

Calibrate thresholds on warmup data and assign regime_{hat}

Output: exogenous table with mid, σ_{hat} , regime_true , regime_{hat}

6.3 Market-Making Environment Step

Input: action (h_idx , m_idx), current simulator state, exogenous row

Decode $h = h_idx + 1$ and $m = m_idx - 2$

Set $\Delta_{bid} = \max(1, h + m)$, $\Delta_{ask} = \max(1, h - m)$

Compute fill intensities and per-step fill probabilities

Sample bid and ask fills

Update inventory and cash, including fees

Advance mid-price and compute W_{t+1}

Return observation, reward = $\Delta_{equity} - \eta * \text{inventory}^2$, done flag, info

6.4 PPO Training and OOS Evaluation Protocol

For each seed and variant:

generate one exogenous synthetic path

split path chronologically into train and test segments

configure observation channels for the variant

```

train PPO on the train segment for the configured timesteps
evaluate the deterministic policy on the test segment
log Sharpe-like ratio, final equity, inventory p99, fill rate, and per-regime metrics
Aggregate seed-paired statistics across variants

```

6.5 WP6 Signal-Informativeness Sweep

For each degradation condition in {full, noisy, lagged, coarsened, none}:

```

    transform sigma_hat according to the condition

```

```

    for each valid variant and seed:

```

```

        train PPO and evaluate OOS

```

Aggregate condition-variant means and paired comparisons

Test whether combined gains value as sigma_hat is degraded

6.6 Reference Implementation Map

Component	Reference file
Run lifecycle	src/run_context.py, run.py
Simulator	src/wp1/sim.py
Regime generation and detectors	src/wp2/synth_regime.py
Gymnasium environment	src/wp3/env.py
PPO training	src/wp4/job_w4_ppo.py
WP5 evaluation	src/wp5/job_w5_eval.py
WP6 sweep	src/wp6/job_w6_sweep_full.py

6.7 Complexity Analysis

Simulation and deterministic evaluation are linear in the number of environment steps. PPO training cost is approximately linear in total timesteps, policy-network forward/backward passes, and the number of seed-variant cells. The WP6 full sweep is expensive because it multiplies conditions, variants, and seeds; this is why its outputs are treated as frozen evidence.

6.8 Executable Example Commands

Commands

```
python run.py --config config/w5_main.json
python run.py --config config/w5_detector_full.json
python run.py --config config/w5_eta_regime.json
python run.py --config config/w5_misspec_mild.json
python run.py --config config/w6_sweep_full.json
python run.py --config config/w6_sweep_full.json --resume <run_id>
```

7 Experimental Setup

7.1 Canonical Configuration Protocol

Canonical experiments are driven by JSON configs under `config/`. Shared parameters include $\text{mid0} = 100.0$, $\text{tick_size} = 0.01$, $\text{dt} = 0.2$, baseline $\text{sigma_mid_ticks} = 0.8$, $A = 5.0$, $k = 1.5$, $\text{fee_bps} = 0.2$, and $\text{latency_steps} = 1$.

7.2 Data Generation and Preprocessing

Each run generates synthetic mid-price and regime paths, computes rolling realized volatility, and assigns detector labels. The main reported pipelines use causal rv_baseline labels.

7.3 Train/Test Split

WP5 and WP6 use a chronological 70/30 split on the exogenous series. PPO trains on the first segment and is evaluated deterministically on the held-out OOS segment.

7.4 Strategy Variants

The main WP5 comparison evaluates naive, AS, ppo_aware , and ppo_blind . The ablation and WP6 designs use sigma_only , regime_only , combined , oracle_pure , and oracle_full .

7.5 Detector Variants

Detector	Role	Defense-safe caveat
rv_baseline	Main causal detector	Rolling RV threshold; used in main WP4/WP5/WP6 pipelines.
rv_dwell	Auxiliary robustness detector	Offline dwell smoothing; not the main causal detector.
hmm	Robustness detector	Higher classification accuracy but no reliable PPO advantage.

7.6 PPO Hyperparameters

Hyperparameter	Canonical full-run value
total_timesteps	1,000,000
learning_rate	3e-4
n_steps	2048
batch_size	256
n_epochs	10
gamma	0.999
clip_range	0.2
ent_coef	0.01 for full WP5/WP6 runs

7.7 Protected Evidence and No-Rerun Policy

The thesis_31 adaptation reuses frozen results. It does not rerun PPO, WP5, WP6, figure scripts, or protected CSV generation. Protected evidence hashes are documented in `EVIDENCE\_MANIFEST.md` and repeated in the reproducibility checklist.

8 Results and Visualisation

8.1 Main WP5 OOS Results

The main 20-seed OOS evaluation shows that PPO variants produce much higher risk-adjusted performance than the fixed-spread and AS baselines. AS can produce higher raw equity, but with substantially larger inventory exposure.

Strategy	Mean Sharpe-like	Mean final equity	Mean inv_p99	Mean fill rate
ppo_aware	0.715	4.10	2.00	0.236
ppo_blind	0.740	4.42	2.05	0.232
AS	0.105	5.05	29.95	0.444
naive	0.127	4.49	21.20	0.119

AS has higher raw equity but carries substantially larger inventory tail risk; therefore the main comparison is risk-adjusted performance and inventory control, not raw equity alone.

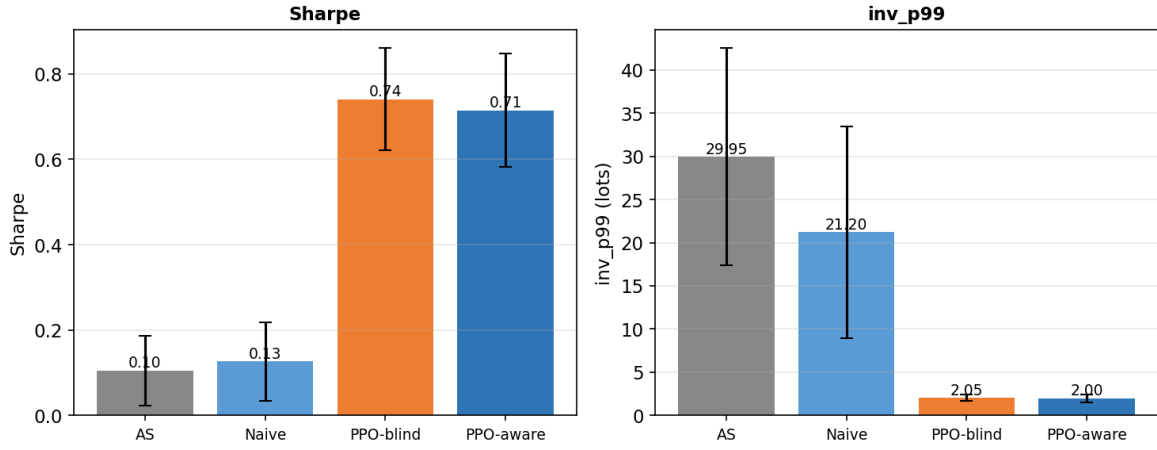


Figure 1. Main OOS performance and inventory risk, reused from frozen thesis figures.

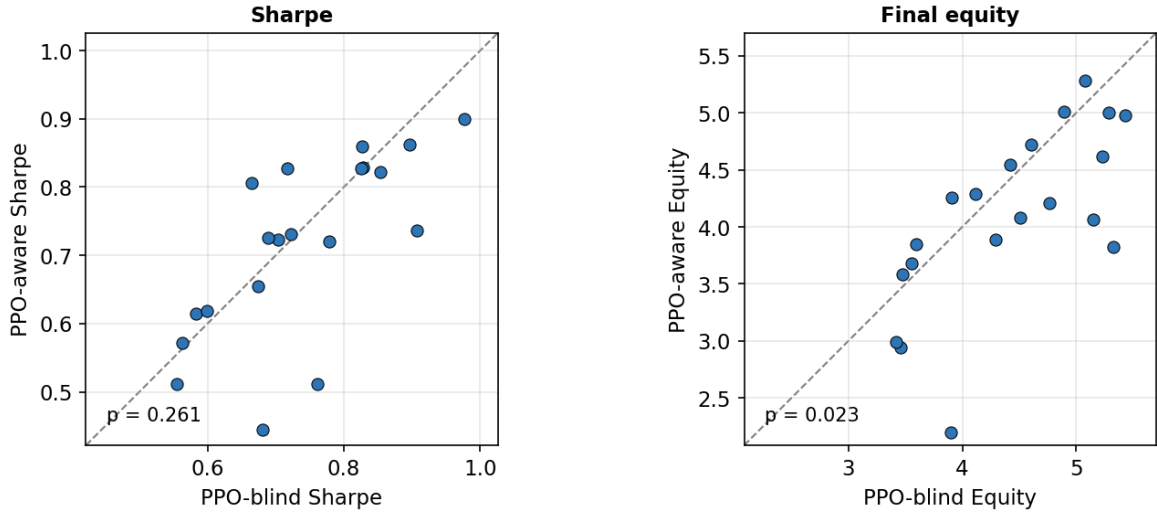


Figure 2. Seed-paired PPO-aware versus PPO-blind comparison.

8.2 Five-Variant Ablation

Variant	Observation meaning	Mean Sharpe-like	Mean final equity	Mean inv_p99
ppo_sigma_only	continuous sigma_hat only	0.752987	4.547923	1.950000
ppo_oracle_full	sigma_hat + true regime one-hot	0.722443	4.067945	2.050000
ppo_regime_only	estimated regime one-hot only	0.697552	4.005379	2.150000
ppo_combined	sigma_hat + estimated regime one-hot	0.696182	3.905214	1.800000

ppo_oracle_pure	true regime one-hot only	0.683829	3.925499	1.950000
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The strongest ablation result is that sigma_only has the highest mean Sharpe-like value, while oracle_full does not significantly beat it. TOST supports practical equivalence under the +/-0.10 Sharpe-like bound.

Comparison	Metric or test	Canonical result	Defense-safe reading
PPO-aware vs PPO-blind	Sharpe-like paired t-test	$p = 0.261$	No significant Sharpe improvement from explicit regime labels.
PPO-aware vs PPO-blind	Final equity paired t-test	$p = 0.023$	Final equity favors PPO-blind.
ppo_sigma_only vs ppo_oracle_full	Sharpe-like paired t-test	$p = 0.115$	No significant oracle-label Sharpe improvement over sigma_hat alone.
ppo_sigma_only vs ppo_oracle_full	TOST +/-0.10	$p = 0.00067$; 90% CI [-0.001, +0.063]	Positive evidence of practical equivalence under the stated bound.
HMM detector	Detector accuracy	81.8%	Higher detector accuracy still does not create a reliable PPO advantage.
Detector robustness	ANOVA	$p = 0.997$	Detector choice does not explain the null result.
Regime-conditional eta	sigma_only vs combined Sharpe	$p = 0.0016$	Result favors sigma_only.
Mild misspecification	sigma_only vs oracle_full Sharpe t-test	$p = 0.881$	No significant Sharpe difference.
Mild misspecification	TOST +/-0.05	$p = 0.042$	Practical equivalence is supported under the stated bound.

The t-test results show no significant Sharpe improvement from explicit regime labels, while the TOST result provides positive evidence of practical equivalence under the stated bound.

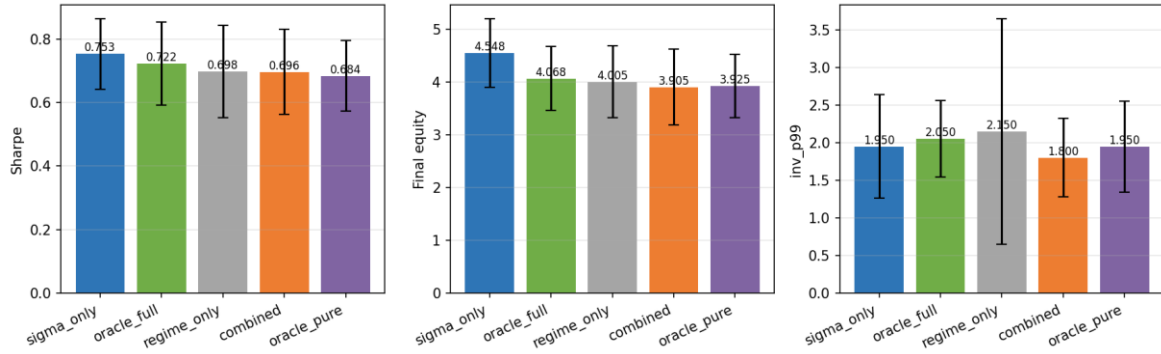


Figure 3. Five-variant ablation summary.

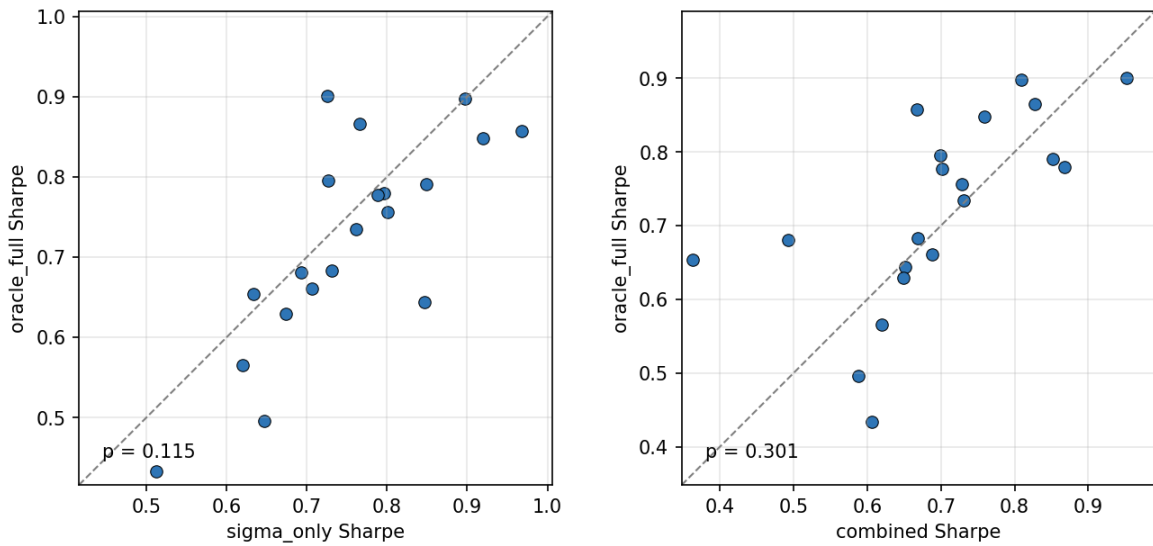


Figure 4. Oracle-label paired-seed comparison.

8.3 Detector Robustness

Detector	Aware vs blind conclusion	Key statistic
rv_baseline	No reliable aware Sharpe advantage	$p = 0.1142$
rv_dwell	No reliable aware Sharpe advantage	$p = 0.1095$
hmm	No reliable aware Sharpe advantage despite higher accuracy	$p = 0.0822$
ANOVA	Detector choice does not explain the result	$F = 0.0034, p = 0.9966$

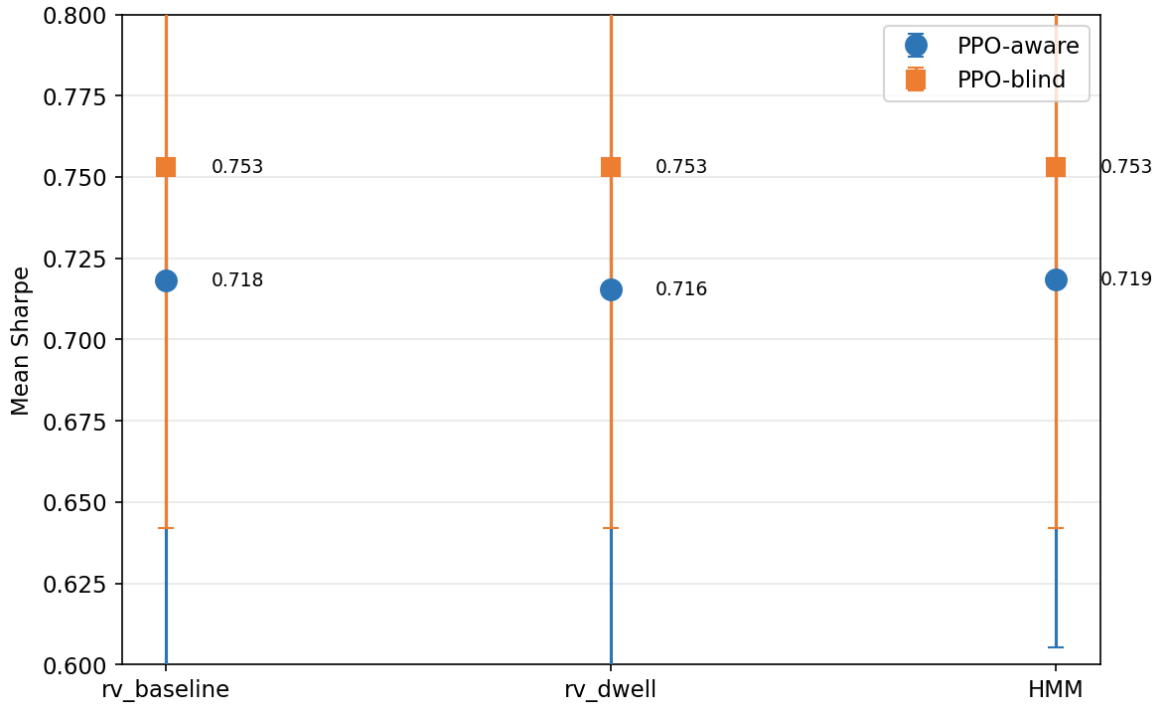


Figure 5. Detector robustness across rv_baseline, rv_dwell, and HMM.

8.4 Reward-Shaping and Misspecification Checks

The regime-conditional eta run tests whether explicit labels become useful when the reward penalizes high-volatility inventory more strongly. sigma_only still beats combined on Sharpe-like performance ($p = 0.0016$). Under mild regime-dependent execution misspecification, sigma_only and oracle_full remain statistically indistinguishable and practically equivalent under the reported TOST bound.

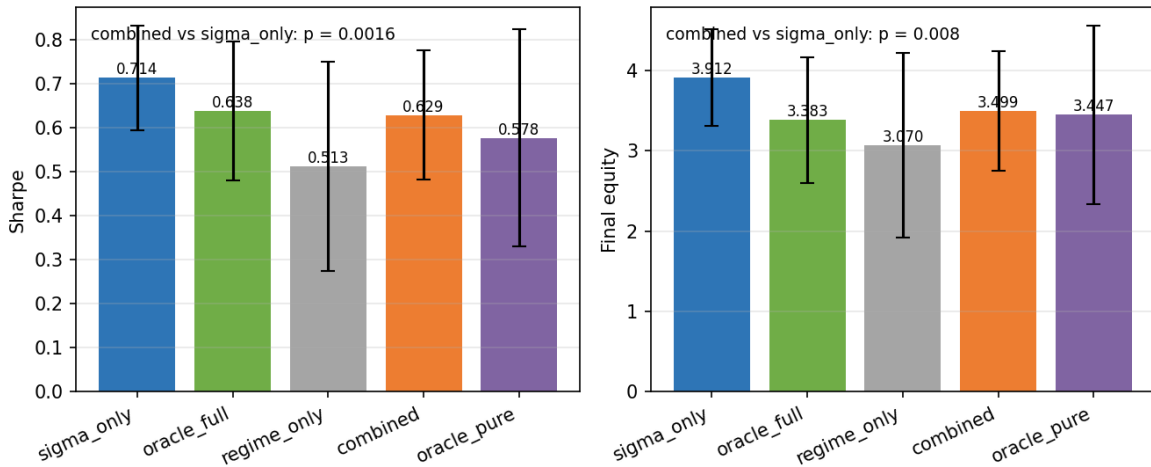


Figure 6. Regime-conditional eta summary.

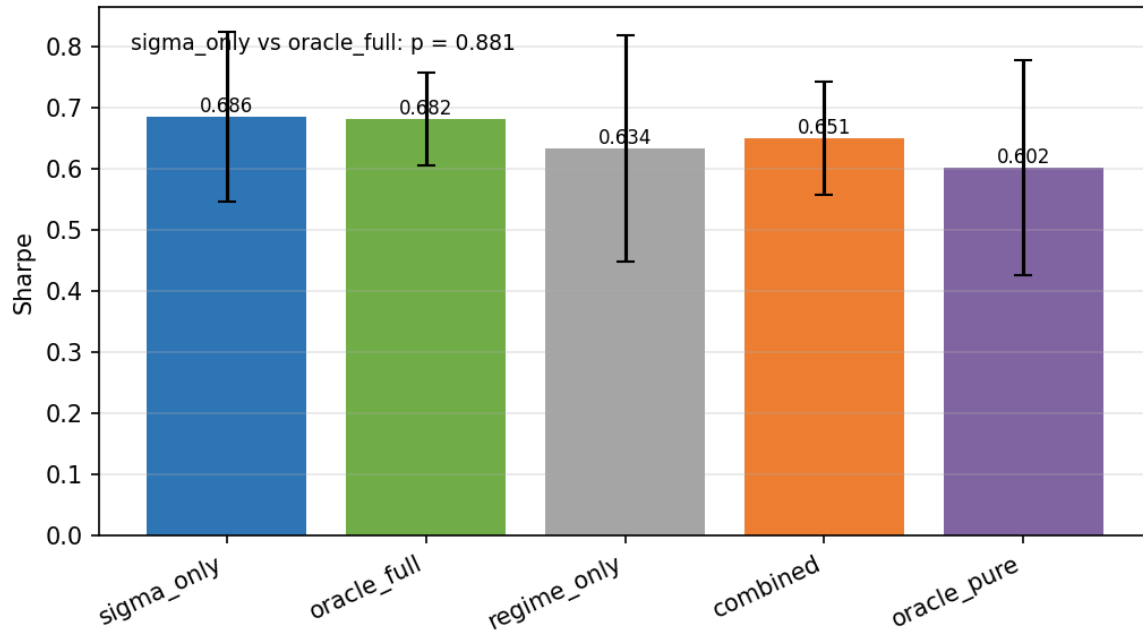


Figure 7. Mild model-misspecification summary.

8.5 WP6 Signal-Informativeness Sweep

condition	sigma_only	combined	regime_only	oracle_full	oracle_pure
full	0.7626 +/- 0.0596	0.6898 +/- 0.0607	0.6991 +/- 0.0668	0.6811 +/- 0.0697	0.6632 +/- 0.0769
noisy	0.7563 +/- 0.0616	0.7125 +/- 0.0628	0.6991 +/- 0.0668	0.6736 +/- 0.1013	0.6632 +/- 0.0769
lagged	0.7822 +/- 0.0454	0.6812 +/- 0.0536	0.6991 +/- 0.0668	0.7264 +/- 0.0686	0.6632 +/- 0.0769
coarsened	0.7827 +/- 0.0533	0.6908 +/- 0.0827	0.6991 +/- 0.0668	0.7447 +/- 0.0549	0.6632 +/- 0.0769
none	undefined	0.6991 +/- 0.0668	0.6991 +/- 0.0668	0.6632 +/- 0.0769	0.6632 +/- 0.0769

WP6 did not support the original informativeness-threshold hypothesis within the tested calibration band. sigma_only remained high under informative conditions, while combined was directionally below sigma_only.

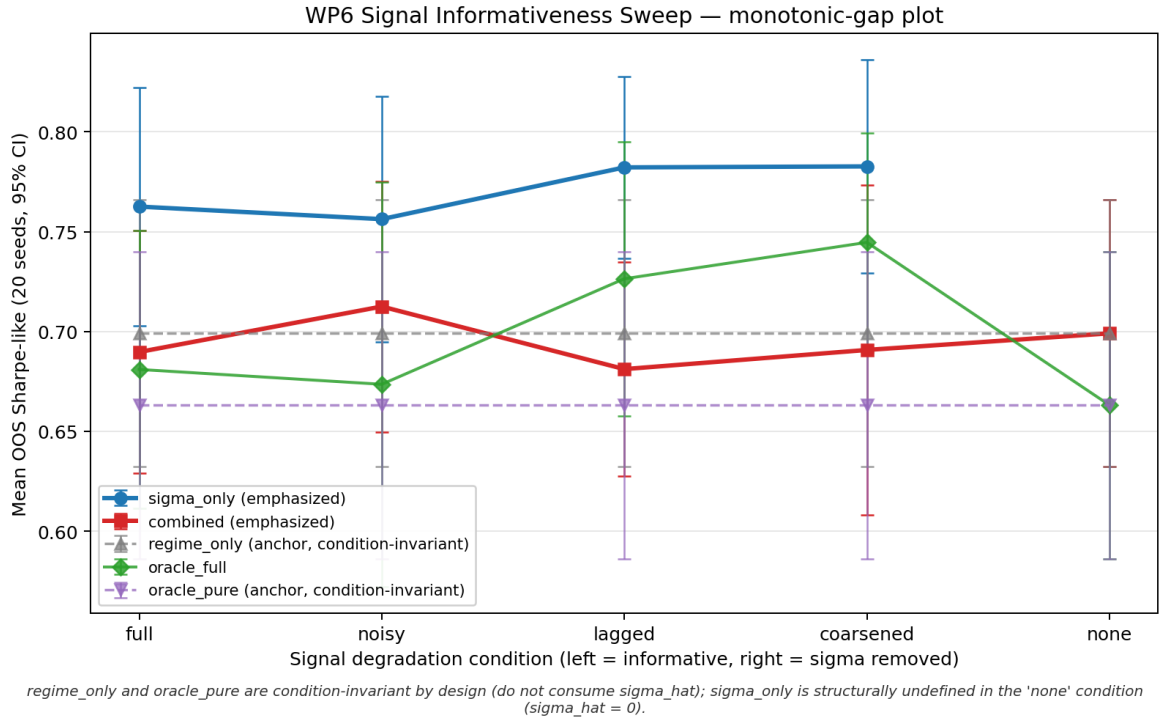
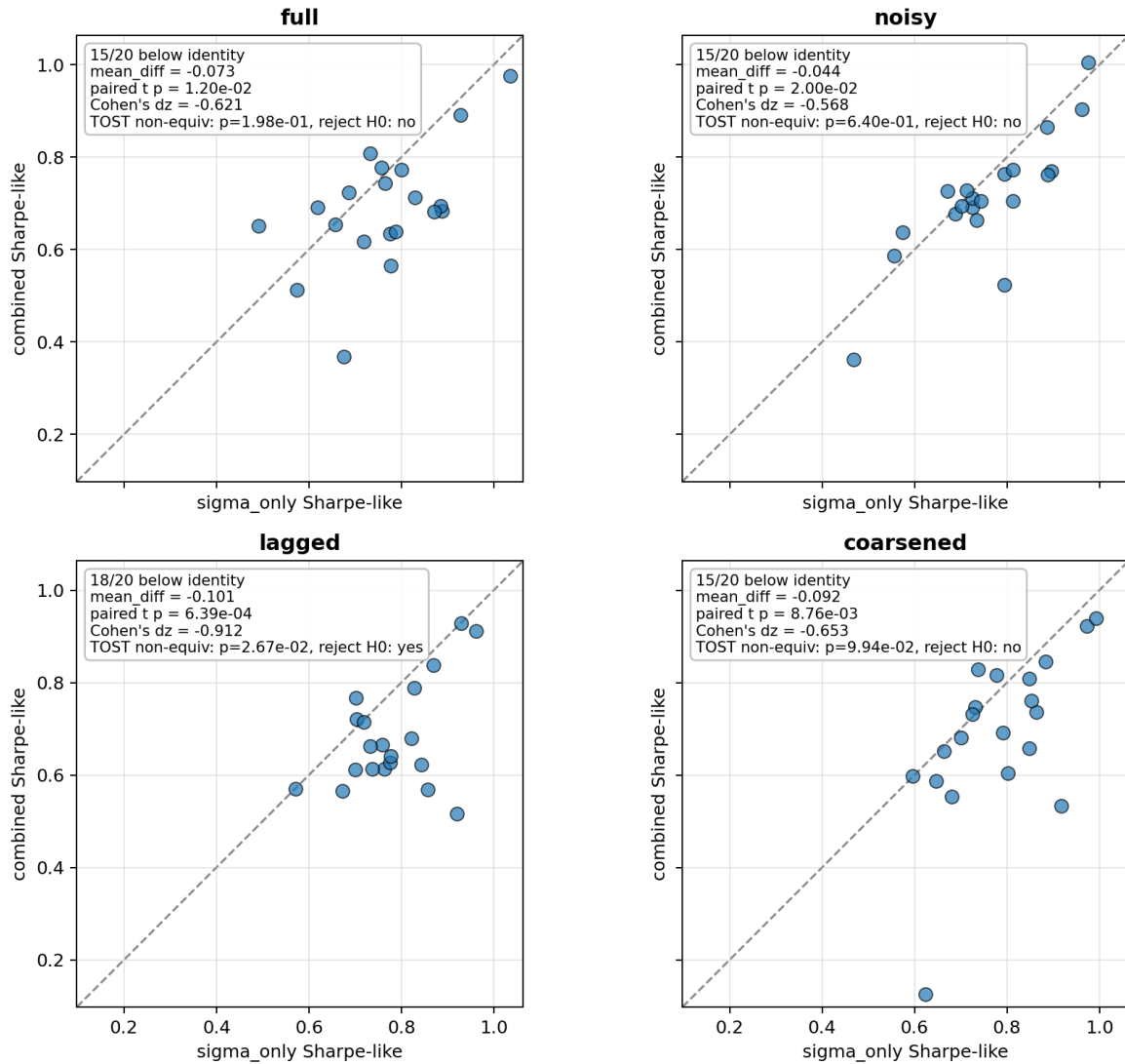


Figure 8. WP6 monotonic-gap plot.

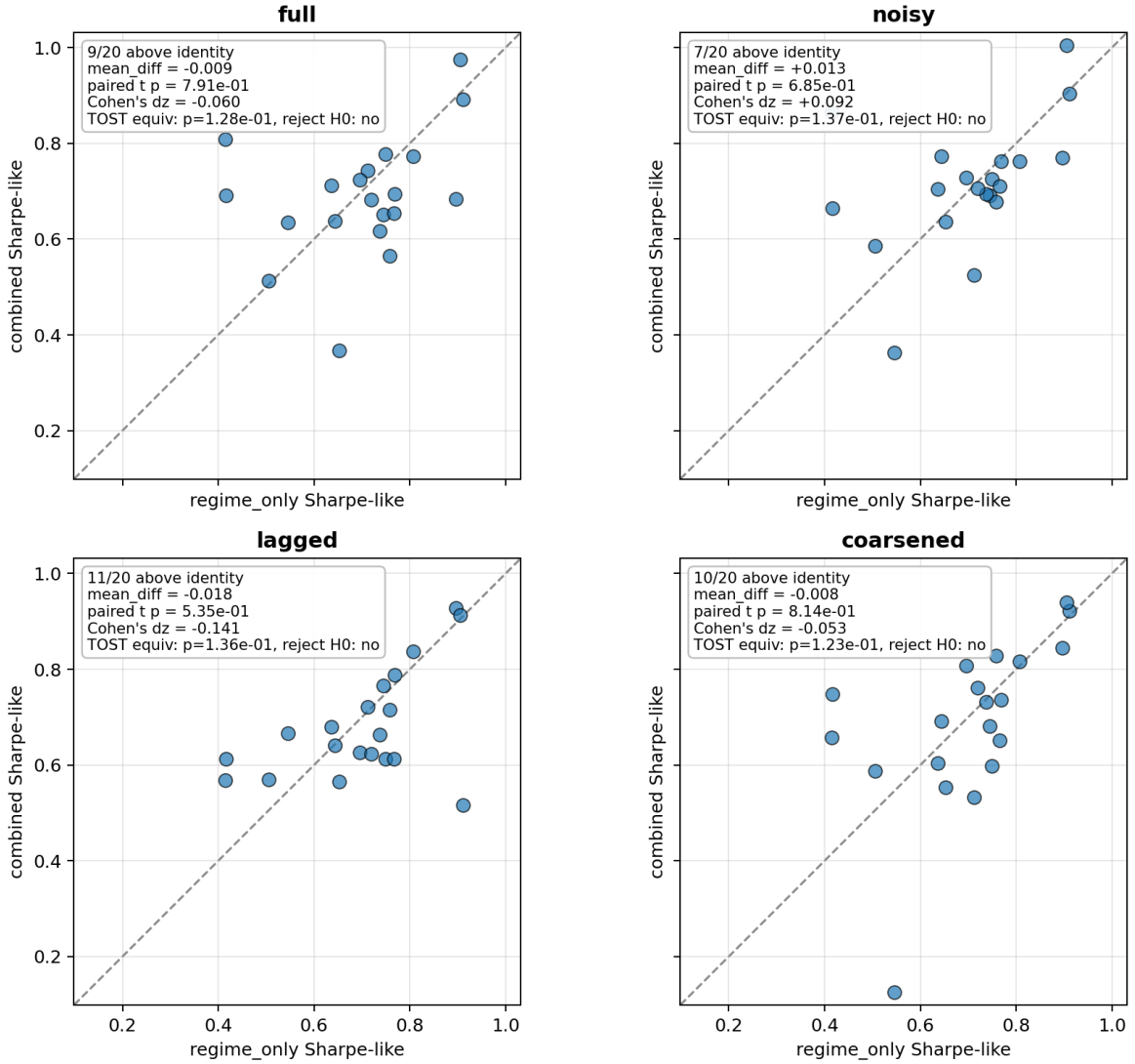
Paired-seed comparison: combined vs sigma_only across conditions



Points below identity line indicate combined underperforms sigma_only for that seed. Delta for TOST = 0.05.

Figure 9. WP6 paired-seed combined versus sigma_only.

Paired-seed comparison: combined vs regime_only across conditions



Points above identity line indicate combined adds value over regime_only for that seed. Delta for TOST = 0.05.

Figure 10. WP6 paired-seed combined versus regime_only.

9 Evaluation Metrics

9.1 Final Equity

Final equity is the terminal marked-to-market wealth. It is economically meaningful but not sufficient alone because high raw equity can be earned by taking large inventory risk.

9.2 Sharpe-Like Ratio

$$\text{Sharpe-like} = \text{mean}(r_t) / \text{std}(r_t)$$

Here r_t denotes the per-step or per-episode return series used by the project metric convention. This is the primary risk-adjusted performance metric.

9.3 Inventory Tail Risk

$$\text{inv_p99} = \text{percentile_99}(|q_t|)$$

Here q_t is inventory. inv_p99 is a practical tail-risk diagnostic for whether a strategy earns returns by carrying large inventory.

9.4 Fill Rate

$$\text{fill_rate} = \text{filled quote opportunities} / \text{total quote opportunities}$$

The numerator and denominator follow the logged project convention. Fill rate is diagnostic rather than the primary thesis metric.

9.5 Paired t-Tests

$$t = \text{mean}(d_i) / (\text{sd}(d_i) / \sqrt{n})$$

Here d_i is the paired seed-level difference between two strategies, $\text{sd}(d_i)$ is its sample standard deviation, and n is the number of paired seeds.

9.6 TOST Equivalence Tests

TOST evaluates whether the plausible range of a paired difference lies inside a pre-specified practical-equivalence band. At $\alpha = 0.05$, the corresponding equivalence reading uses the 90% confidence interval. WP5 uses TOST to support practical equivalence for sigma_only versus oracle_full ; WP6 also uses related paired tests to examine non-equivalence and mean indistinguishability patterns.

10 Discussion

10.1 Main Interpretation: Signal Redundancy

In the tested controlled synthetic HFMM environment, explicit categorical volatility-regime labels do not provide robust incremental value once sigma_hat is already observed by the PPO policy. The results are consistent with signal redundancy.

The interpretation is that the continuous realized-volatility proxy already supplies the economically useful volatility information needed by the policy in this environment. The categorical label may be coarser or redundant when presented alongside sigma_hat .

10.2 Why the Result Is Not a Weak Null

The conclusion does not rest only on $p > 0.05$. It is supported by detector robustness, oracle labels, TOST equivalence, reward-channel checks, mild misspecification, and WP6 degradation tests.

10.3 Synthetic-Market Boundary

The result is bounded to the tested synthetic HFMM simulator. It does not imply that volatility-regime structure lacks importance in finance or that categorical signals cannot help in other market-making designs.

10.4 Risks to Interpretation

Risk	Mitigation
Overclaiming mechanism	Use signal-redundancy and categorical-channel degradation as interpretations, not proofs of PPO internals.
Detector causality confusion	State that <code>rv_baseline</code> is the main causal detector and <code>rv_dwell</code> is auxiliary/offline.
Synthetic external validity	Frame all claims as controlled synthetic-market claims.
Template over-compression	Keep the literature and evidence spine, but map it into template headings.

10.5 Future Work

- Evaluate stronger regime-dependent execution misspecification.
- Test real or richer limit-order-book data after thesis scope, not as a claim of this report.
- Run representation-level diagnostics only as future mechanism work.
- Explore alternative observation encodings and policy architectures.

11 Reproducibility Checklist

Item	Status in thesis_31
Frozen baseline	thesis-v29-frozen tag; commit 9681faa; thesis_29 remains untouched.
Template draft	thesis_31 is a new template-adapted draft, not a replacement for thesis_29 or thesis_30.
Run ID structure	YYYYMMDD-HHMMSS_seed<S>_<TAG>_<COMMIT>.
Config snapshots	Every run snapshots config; config_snapshot_all.md inventories active config files.
Seeds	WP5 main uses seeds 1-20; WP6 full uses seeds 42-61.
Train/test split	70/30 chronological split on exogenous synthetic series.
Evidence manifest	EVIDENCE_MANIFEST.md records protected evidence and remediation invariants.

Protected CSV hashes	Listed below and verified by file hash checks.
Codebase snapshot	docs/internal/codebase_snapshot.py.
No-rerun policy	No PPO/WP5/WP6 reruns or figure regeneration during template adaptation.

Protected artifact	Expected SHA256	Check
results/metrics_detector_compare.csv	28E7AD40BB47...	MATCH
docs/internal/wp6_sweep_full/summary_condition_variant.csv	6DD627E81637...	MATCH
docs/internal/wp6_sweep_full/summary_paired_combined_vs_sigma.csv	4BABCAAACE1D...	MATCH
docs/internal/wp6_sweep_full/summary_paired_combined_vs_regime.csv	2087FEFBE5DC...	MATCH

Reproducibility command	Purpose
python -m venv .venv	Create environment.
& .\venv\Scripts\Activate.ps1	Activate environment on Windows PowerShell.
pip install -r requirements.txt	Install dependencies.
python run.py --config config/w5_main.json	Run main WP5 evaluation.
python run.py --config config/w6_sweep_full.json	Run WP6 sweep.
python run.py --config config/w6_sweep_full.json --resume <run_id>	Resume WP6 sweep.
ruff check src/	Lint source code.

These commands document reproducibility paths only; thesis_31 adaptation did not rerun experiments.

12 Conclusion

This template-adapted report preserves the thesis finding while reorganizing the manuscript into the requested technical-report structure. PPO learns strong risk-adjusted quoting behavior in the synthetic HFMM simulator, but the explicit categorical regime channel does not provide robust incremental value

beyond σ_{hat} . The result is best defended as evidence consistent with signal redundancy in the tested controlled environment.

Appendix A Code Appendix

Area	Files
Run management	run.py; src/run_context.py
Simulation	src/wp1/sim.py; src/wp1/w1_as_baseline.py
Regimes	src/wp2/synth_regime.py; src/wp2/job_w2_synth.py
Environment	src/wp3/env.py; src/wp3/w3_sanity.py
PPO training	src/wp4/job_w4_ppo.py
Evaluation	src/wp5/job_w5_eval.py; src/wp5/job_w5_detector_compare.py
Signal audit and sweep	src/wp5_5/*; src/wp6/job_w6_sweep_full.py

Canonical commands

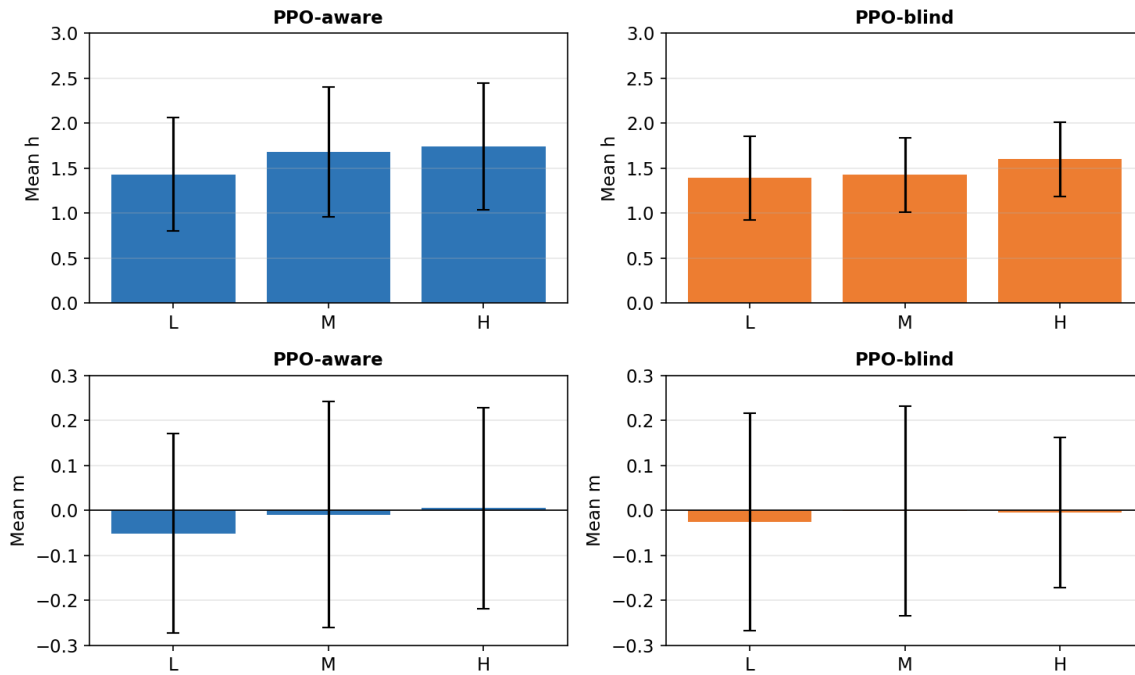
```
python run.py --config config/w5_main.json
python run.py --config config/w5_detector_full.json
python run.py --config config/w5_eta_regime.json
python run.py --config config/w5_misspec_mild.json
python run.py --config config/w6_sweep_full.json
```

Appendix B Sanity Checks and Unit Tests

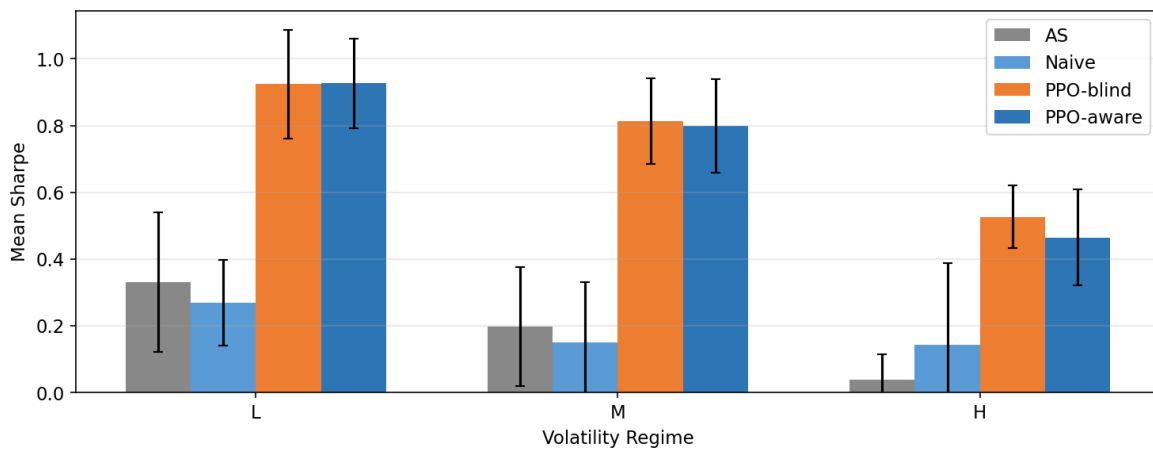
Check	Purpose
tests/test_csv_metric_logger.py	Verifies stable CSV metric schema behavior.
tests/test_resume_validation.py	Verifies resume-mode config validation.
WP3 sanity checks	Exercises naive, AS, and random policies through the Gymnasium environment.
Figure/path checks	Ensures thesis_31 references existing frozen figure files.
Forbidden phrase scan	Guards against overclaims in the generated draft.

Appendix C Extended Evidence Tables

This appendix contains diagnostic material demoted from the main body to keep the report template focused.



Appendix Figure C1. Regime-wise action distribution diagnostic.



Appendix Figure C2. Regime-wise Sharpe diagnostic.

Diagnostic	Status	Interpretation
Post-hoc signal diagnostics	Supporting only	Consistent with signal redundancy but not primary

		evidence.
WP5.5 signal audit	Offline calibration	Supports WP6 degradation design; no PPO training.
Detector robustness statistics	Protected evidence	Addresses poor-detector objection without changing main detector caveat.
WP6 paired summaries	Protected evidence	Supports refined interpretation of the signal-informativeness sweep.